

Microsoft Sentinel

Cloud-native SIEM + SOAR built on Azure Log Analytics for detection, investigation, and response

Overview

Microsoft Sentinel is a scalable, cloud-native SIEM and SOAR solution that ingests telemetry from Microsoft and third-party sources into a Log Analytics workspace, applies analytics rules and UEBA/ML detections, and enables automated response through Playbooks (Logic Apps). It underpins most modern Microsoft-centric SOC operations.

Core Architecture Components

	Role
Log Analytics Workspace	Backing data store; KQL query engine over ingested tables
Data Connectors	Native connectors (Defender suite, Entra ID, AWS, GCP, syslog/CEF) or Codeless Connector Platform
Analytics Rules	Scheduled/near-real-time/ML rules that generate Incidents from suspicious patterns
Workbooks	Interactive dashboards for hunting and reporting
Playbooks	Logic Apps that automate investigation/response (isolate host, disable user, notify)
Watchlists	Curated reference lists (VIP users, known-bad IPs) used to enrich/filter detections
Content Hub / Solutions	Packaged analytics rules, workbooks, connectors per product (e.g., "Microsoft Defender for Cloud")

Analytics Rule Types

- **Scheduled** — KQL query run on an interval, most common custom detection type
- **NRT (Near Real Time)** — sub-minute latency for time-critical detections

- **Microsoft Security** — auto-imports high-fidelity alerts from Defender products as Incidents
- **Fusion** — ML-based multi-stage attack correlation across signal sources
- **ML Behavior Analytics** — built-in UEBA anomaly detection for identity/entity behavior

Investigation Workflow

Incident (grouped alerts + entities) → Investigation graph (pivots across related entities: user, host, IP, file) → Entity behavior timeline → Hunting queries (KQL) to validate scope → Playbook-driven containment → Incident closed with classification (True/False Positive, reason).

KQL Essentials

	Use
<code>where</code>	Filter rows by condition
<code>summarize</code>	Aggregate (count, dcount, arg_max) grouped by field
<code>join</code>	Correlate across tables (e.g., SigninLogs with AuditLogs)
<code>bin()</code>	Time-bucket for trend/anomaly analysis
<code>extend</code>	Add computed columns
<code>project</code>	Select/rename output columns

Cost & Ingestion Management

- Pricing driven primarily by **GB ingested/day** — commitment tiers reduce per-GB cost at scale
- **Basic Logs / Auxiliary Logs** tier for high-volume, low-query-frequency data (cheaper ingestion, limited KQL)
- **Data Collection Rules (DCR)** filter/transform data before ingestion to reduce noise and cost
- Set **table-level retention** independently from workspace default to control storage cost

Common Pitfalls

Fix

Ingesting all logs at verbose level by default	Use DCR filtering/transformation and Basic Logs tier for high-volume, low-value tables
Analytics rules with no entity mapping	Map entities (account, host, IP) so incidents correlate and drive UEBA/investigation graphs
Alert fatigue from unfused, low-fidelity rules	Tune thresholds, use Fusion, and suppress known-benign patterns via watchlists
Playbooks with no error handling	Add try/catch branches in Logic Apps; alert if automated response itself fails
No retention strategy per table	Set table-level retention distinct from workspace default to balance cost and investigation needs

Interview / Exam Quick Hits

- Sentinel is **Log Analytics + security content layer**, not a separate data store
- Fusion detections correlate **low-fidelity signals** into a single high-confidence multistage incident
- Playbooks are triggered manually, automatically on incident creation, or on a schedule
- UEBA baselines normal behavior per entity over ~time window to flag deviations (impossible travel, anomalous sign-in)
- MITRE ATT&CK mapping is built into analytics rules for coverage-gap analysis via the MITRE workbook